1 SC - Small Cancellation Group

1.1 Mathematical Description

Groups with presentation $\langle a, b \mid r_1, r_2, \ldots \rangle$ satisfying the C'(1/6) small cancellation condition. This condition ensures that any common prefix between two distinct relators (or a relator and its cyclic conjugate) has length less than 1/6 of either relator. The word problem is solvable in $O(n)$ time via Dehn's algorithm.

Example Relator: $r = ababab ABABAB$ (length 12)

1.2 Alphabet

Symbol	Meaning
`a`	Generator a
`A`	Inverse a^{-1}
`b`	Generator b
`B`	Inverse b^{-1}

1.3 Identity Test

Dehn's Algorithm: Scan the word for any subword matching more than half of a cyclic conjugate of a relator. Replace with the shorter complement. Repeat until no replacements possible. Identity iff reduces to empty.

1.4 Dataset Variants

1.4.1 Synthetic (sc_synthetic_100k.csv)

Word	Label	Length
`aAbbBBaAbBAb`	0	12
`BababABABABb`	1	14
`aAbBababABABAB`	1	16

1.4.2 Augmented (sc_augmented_100k.csv)

Word	Label	Length
`AAbabaBBaabBAbabA`	0	17
`abababABABABaAbbBB`	1	18
`BAabababABABABAbbaAB`	1	20

1.4.3 Hard (sc_hard_100k.csv)

Word	Label	Length
`AAbaBAbabABaBAbaBAbabBAbaB`	0	26
`abababABABABBabababABABABbAaBB`	1	30
`AABBabababABABABaabbABabababABABABba`	1	34

2 TF - Thompson's Group F

2.1 Mathematical Description

Thompson's group F consists of piecewise-linear homeomorphisms of $[0, 1]$ with dyadic rational breakpoints and slopes that are powers of 2. Infinite presentation:

$\langle x_0, x_1, x_2, \ldots \mid x_k^{-1} x_n x_k = x_{n+1}$ **for** $k < n \rangle$

Using finite generators x_0, x_1, the key derived relation is $X_1 x_0 x_1 = x_0 x_0$.

2.2 Alphabet

Symbol	Meaning
`x0`	Generator x_0
`X0`	Inverse x_0^{-1}
`x1`	Generator x_1
`X1`	Inverse x_1^{-1}

2.3 Identity Test

Tree Pair Normalization: Each element represented by binary tree pair. Identity iff normalized trees are identical.

2.4 Dataset Variants

2.4.1 Synthetic (tf_synthetic_100k.csv)

Word	Label	Length
`x0 x1 X0 x0 X1`	0	5
`x1 X1 x0 X0`	1	4
`X1 x0 x1 X0 X0`	1	5

2.4.2 Augmented (tf_augmented_100k.csv)

Word	Label	Length
`x0 x1 x0 X1 x1 X0 X0 x0 X1 x0 x1`	0	11
`x0 X1 x0 x1 X0 X0 X0 x1 X1 x0`	1	10
`x1 x0 X1 x0 x1 X0 X0 x0 X0 x1 X1`	1	11

2.4.3 Hard (tf_hard_100k.csv)

Word	Label	Length
`x0 x0 x1 X0 x1 x0 X1 X0 x0 x1 x0 X1 X0 x1 x0 X0 X1 x0 x1 X0 x0`	0	21
`X1 x0 x1 X0 X0 x1 X1 x0 x1 X0 X0 x0 X0 X1 x0 x1 X0 X0 x1 X1`	1	20
`x0 X1 x0 x1 X0 X0 x1 x1 X1 X1 x0 X0 X1 x0 x1 X0 X0 x0 x0 X0 X0`	1	21

3 H3 - Heisenberg Group $H_3(\mathbb{Z})$

3.1 Mathematical Description

The discrete Heisenberg group of 3×3 upper triangular integer matrices:

$$\begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

Presentation: $\langle a, b, c \mid [a, b] = c, [a, c] = 1, [b, c] = 1 \rangle$
Critical: $abAB = c \neq$ identity (the commutator equals c, not 1!)

3.2 Alphabet

Symbol	Meaning
`a`	Generator a
`A`	Inverse a^{-1}
`b`	Generator b
`B`	Inverse b^{-1}
`c`	Central element $c = [a, b]$
`C`	Inverse c^{-1}

3.3 Identity Test

Normal Form Collection: Rewrite to $a^x b^y c^z$ form. Identity iff $x = y = z = 0$.

3.4 Dataset Variants

3.4.1 Synthetic (h3_synthetic_100k.csv)

Word	Label	Length
`a b c A`	0	4
`a b A B C`	1	5
`b a B A c`	1	5

3.4.2 Augmented (h3_augmented_100k.csv)

Word	Label	Length
`a a b c c A B C A`	0	10
`c a b A B C c C b B`	1	10
`a b A B C b a B A c`	1	10

3.4.3 Hard (h3_hard_100k.csv)

Word	Label	Length
`a b a b c A A B B c c C C a b A B`	0	18
`a b A B C a b A B C b a B A c b a B A c`	1	20
`c c a b A B C C a a b b A A B B C C c c a b A B C C`	1	26

4 BS12 - Baumslag-Solitar Group BS(1,2)

4.1 Mathematical Description

BS(1,2) $= \langle a, t \mid t^{-1}at = a^2 \rangle$

A fundamental **non-automatic group**. The relation states conjugating a by t doubles it: $Tat = aa$.

4.2 Alphabet

Symbol	Meaning
`a`	Generator a
`A`	Inverse a^{-1}
`t`	Generator t
`T`	Inverse t^{-1}

4.3 Identity Test

Britton's Lemma: Apply $Tat \to aa$, reduce to normal form. Note: `t a T` does NOT reduce (would require $a^{1/2}$).

4.4 Dataset Variants

4.4.1 Synthetic (bs12_synthetic_100k.csv)

Word	Label	Length
`t a T a`	0	4
`T a t A A`	1	5
`a T a t A A A`	1	7

4.4.2 Augmented (bs12_augmented_100k.csv)

Word	Label	Length
`t t a a T T a a a`	0	9
`T a a t A A A A t T`	1	10
`a a T a t A A A A A`	1	10

4.4.3 Hard (bs12_hard_100k.csv)

Word	Label	Length
`t T a a t a T T a a a a T a t A A`	0	17
`T T a t t A A A A T a a t A A A A t T`	1	19
`a T a t A A A T a t A A A A A A a A t T`	1	20

5 OR - One-Relator Group

5.1 Mathematical Description

Groups with a single defining relator. We use the classic example:

$\langle a, b \mid abab = baba \rangle$

This is isomorphic to the trefoil knot group. The relation gives: $ababBABA =$ identity.

5.2 Alphabet

Symbol	Meaning
a	Generator a
A	Inverse a^{-1}
b	Generator b
B	Inverse b^{-1}

5.3 Identity Test

Magnus Algorithm: Apply the relator $ababBABA = 1$ and its cyclic conjugates. Word equals identity iff it reduces to empty via relator applications and free reductions.

5.4 Dataset Variants

5.4.1 Synthetic (or_synthetic_100k.csv)

Word	Label	Length
a b a A b B	0	6
a b a b B A B A	1	8
B a b a b B A B A b	1	10

5.4.2 Augmented (or_augmented_100k.csv)

Word	Label	Length
a a b b a b A A B B	0	10
a A b B a b a b B A B A	1	12
A a b a b B A B A b B a A	1	13

5.4.3 Hard (or_hard_100k.csv)

Word	Label	Length
a b a b a b A B A B A a b a b B A B	0	18
a b a b B A B A B a b a b B A B A b a A	1	20
A a b a b B A B A b B b a b a b B A B A B a A	1	23

6 GG - Grigorchuk Group

6.1 Mathematical Description

The first Grigorchuk group is a self-similar group acting on the infinite binary tree. It is:

- Finitely generated but not finitely presented
- Has intermediate growth (between polynomial and exponential)
- A famous counterexample in geometric group theory

Generators: a, b, c, d where **ALL are self-inverse** ($a^2 = b^2 = c^2 = d^2 = 1$)
Key Relation: $bcd = 1$ (equivalently: $d = bc$, $c = bd$, $b = cd$)

6.2 Alphabet

Symbol	Meaning
a	Generator a (order 2)
b	Generator b (order 2)
c	Generator c (order 2)
d	Generator d (order 2)

Note: No inverse symbols needed since each generator is its own inverse.

6.3 Identity Test

Recursive Automaton: Elements represented by states of a finite automaton acting on binary tree. Identity verified by recursive decomposition.

Simple Identities: aa, bb, cc, dd, bcd, dcb, $aabcd$, etc.

6.4 Dataset Variants

6.4.1 Synthetic (gg_synthetic_100k.csv)

Word	Label	Length
a b c a	0	4
a a b b	1	4
b c d a a	1	5

6.4.2 Augmented (gg_augmented_100k.csv)

Word	Label	Length
a b c d a b c a	0	8
a b c d a a d c b a	1	10
b c d b c d a a c c	1	10

6.4.3 Hard (gg_hard_100k.csv)

Word	Label	Length
a b a c a d b c d a b c a d c b a	0	17
a b c d a a b c d a a b c d a a d c b a	1	20
b c d d c b a a b c d d c b c c b c d d c b	1	22

7 MCG - Mapping Class Group

7.1 Mathematical Description

The mapping class group $\mathrm{MCG}(\Sigma_2)$ of a genus-2 surface, generated by Dehn twists around essential curves.

Generators: Five Dehn twists $\tau_1, \tau_2, \tau_3, \tau_4, \tau_5$

Relations:

- Braid relation: $\tau_i\tau_j\tau_i = \tau_j\tau_i\tau_j$ when $|i-j| = 1$
- Commutation: $\tau_i\tau_j = \tau_j\tau_i$ when $|i-j| \geq 2$

7.2 Alphabet

Symbol	Meaning
`t1`	Dehn twist τ_1
`T1`	Inverse τ_1^{-1}
`t2`	Dehn twist τ_2
`T2`	Inverse τ_2^{-1}
`t3`	Dehn twist τ_3
`T3`	Inverse τ_3^{-1}
`t4`	Dehn twist τ_4
`T4`	Inverse τ_4^{-1}
`t5`	Dehn twist τ_5
`T5`	Inverse τ_5^{-1}

7.3 Identity Test

Curve Action: Verify element acts trivially on all curves, or reduce using braid/commutation relations to normal form.

7.4 Dataset Variants

7.4.1 Synthetic (mcg_synthetic_100k.csv)

Word	Label	Length
`t1 t2 t3 T1 T3`	0	5
`t1 t2 t1 T2 T1 T2`	1	6
`t1 T1 t3 t4 t3 T4 T3 T4`	1	8

7.4.2 Augmented (mcg_augmented_100k.csv)

Word	Label	Length
`t1 t2 t3 t4 t5 T1 T2 T3 T5 t4`	0	10
`t2 t3 t2 T3 T2 T3 t1 T1 t4 T4`	1	10
`t1 t3 t3 T3 T3 t2 t1 T2 T1 T2 t2 T1`	1	12

7.4.3 Hard (mcg_hard_100k.csv)

Word	Label	Length
`t1 t2 t1 t3 t4 t3 T2 T1 T4 T3 t5 t4 t5 T4 T5 T4 t2 t1 T3`	0	19
`t1 t2 t1 T2 T1 T2 t3 t4 t3 T4 T3 T4 t2 t3 t2 T3 T2 T3 t1 T1`	1	20
`t1 t2 t1 T2 T1 T2 t4 t5 t4 T5 T4 T5 t2 t1 t2 T1 T2 T1 t3 T3 t5 T5`	1	22

8 LL - Lamplighter Group

8.1 Mathematical Description

The lamplighter group $\mathbb{Z}_2 \wr \mathbb{Z}$ (wreath product) models a lamplighter walking on an integer line with binary lamps.

Presentation: $\langle a, t \mid a^2 = 1, [a, t^n a t^{-n}] = 1 \text{ for all } n \rangle$
Intuition:

- t moves the lamplighter right, T moves left
- a toggles the lamp at current position
- a is self-inverse ($a^2 = 1$)

8.2 Alphabet

Symbol	Meaning
`a`	Toggle lamp (order 2, $a = a^{-1}$)
`t`	Move right
`T`	Move left (t^{-1})

8.3 Identity Test

Configuration Check: Word equals identity iff:

1. Final position = 0 (equal t's and T's)
2. All lamps off (even toggles at each position)

Examples:

- `t a T a` = NOT identity (lamps at positions 0 and 1 are on)
- `t a t T a T` = identity (lamp 1 toggled twice, end at position 0)

8.4 Dataset Variants

8.4.1 Synthetic (ll_synthetic_100k.csv)

Word	Label	Length
`t a t T T a`	0	6
`a a t T`	1	4
`t a t T a T`	1	6

8.4.2 Augmented (ll_augmented_100k.csv)

Word	Label	Length
`t t a T a t T T a`	0	9
`t t a a T T a a t T`	1	10
`t a t a T a T a T t T`	1	11

8.4.3 Hard (ll_hard_100k.csv)

Word	Label	Length
`t t t a T T a t a T T t a t T a T a t T`	0	20
`t t a t T a t T T a a t a T a T T a T t t T`	1	22
`t a t a t a T T a T a t t T a T T a t a T a T t T`	1	25

9 KT - Knot Group (Trefoil)

9.1 Mathematical Description

The fundamental group of the complement of the trefoil knot in S^3.

Wirtinger Presentation: $\langle a, b \mid aba = bab \rangle$

This is equivalent to the braid group relation and the one-relator group with relator $ababBABA$.

9.2 Alphabet

Symbol	Meaning
`a`	Generator a
`A`	Inverse a^{-1}
`b`	Generator b
`B`	Inverse b^{-1}

9.3 Identity Test

Relator Application: From $aba = bab$, derive $aba \cdot B \cdot A \cdot B = 1$. Apply relator and free reductions.

9.4 Dataset Variants

9.4.1 Synthetic (kt_synthetic_100k.csv)

Word	Label	Length
`a b a A B`	0	5
`a b a B A B`	1	6
`B a b a B A B b`	1	8

9.4.2 Augmented (kt_augmented_100k.csv)

Word	Label	Length
`a b a b A B A a B`	0	9
`a A b B a b a B A B`	1	10
`A a b a B A B b B a A`	1	11

9.4.3 Hard (kt_hard_100k.csv)

Word	Label	Length
`a b a b a B A B A a b a B A B b A B a A`	0	20
`a b a B A B B a b a B A B b a A b B a A`	1	20
`A a b a B A B b B a b a B A B b a b a B A B B A`	1	24

10 SL2Z - Modular Group

10.1 Mathematical Description

The modular group $\mathrm{PSL}(2, \mathbb{Z}) \cong \mathrm{SL}(2, \mathbb{Z})/\{\pm I\}$ is fundamental in number theory and hyperbolic geometry.

Presentation: $\langle S, T \mid S^2 = (ST)^3 = 1 \rangle$ for $\mathrm{PSL}(2, \mathbb{Z})$

For $\mathrm{SL}(2,\mathbb{Z})$: $\langle S, T \mid S^4 = 1, (ST)^3 = S^2 \rangle$
Matrix Representatives:

- $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation by $\pi/2$)
- $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (shear)

10.2 Alphabet

Symbol	Meaning
`S`	Generator S (order 4)
`s`	Inverse $S^{-1} = S^3$
`T`	Generator T
`t`	Inverse T^{-1}

10.3 Identity Test

Matrix Computation: Multiply matrices, check if result is identity matrix I.
Key Relations:

- $SSSS = I$ ($S^4 = 1$)
- $STSTST = SS$ (i.e., $(ST)^3 = S^2$)
- $STSTSTss = I$

10.4 Dataset Variants

10.4.1 Synthetic (sl2z_synthetic_100k.csv)

Word	Label	Length
`S T s T s`	0	5
`S S S S`	1	4
`S T S T S T s s`	1	8

10.4.2 Augmented (sl2z_augmented_100k.csv)

Word	Label	Length
`S T S T s T s T S s`	0	10
`S S S S T t S S S S`	1	10
`T S T S T S T s s t S S S S`	1	14

10.4.3 Hard (sl2z_hard_100k.csv)

Word	Label	Length
`S T S T S T s S T S T s T s T S s s T t S`	0	21
`S T S T S T s s S S S S T t S T S T S T s s`	1	22
`T S T S T S T s s t S S S S S T S T S T s s S S`	1	24

11 Summary Statistics

11.1 Dataset Overview by Group

Group	Variant	Rows	Identity %	Non-Identity %	Length Range	Mean Length
SC	Synthetic	100,000	50.0%	50.0%	4 – 50	27.3
SC	Augmented	100,000	50.0%	50.0%	10 – 50	29.8
SC	Hard	100,000	50.0%	50.0%	20 – 80	48.5
TF	Synthetic	100,000	50.0%	50.0%	4 – 50	26.1
TF	Augmented	100,000	50.0%	50.0%	10 – 50	28.9
TF	Hard	100,000	50.0%	50.0%	20 – 80	47.2
H3	Synthetic	100,000	50.0%	50.0%	4 – 50	25.8
H3	Augmented	100,000	50.0%	50.0%	10 – 50	29.2
H3	Hard	100,000	50.0%	50.0%	20 – 80	46.3
BS12	Synthetic	100,000	50.0%	50.0%	4 – 50	26.4
BS12	Augmented	100,000	50.0%	50.0%	10 – 50	28.7
BS12	Hard	100,000	50.0%	50.0%	20 – 80	48.1
OR	Synthetic	100,000	50.0%	50.0%	4 – 50	26.8
OR	Augmented	100,000	50.0%	50.0%	10 – 50	29.1
OR	Hard	100,000	50.0%	50.0%	20 – 80	47.6
GG	Synthetic	100,000	50.0%	50.0%	4 – 50	25.4
GG	Augmented	100,000	50.0%	50.0%	10 – 50	28.3
GG	Hard	100,000	50.0%	50.0%	20 – 80	46.8
MCG	Synthetic	100,000	50.0%	50.0%	4 – 50	27.1
MCG	Augmented	100,000	50.0%	50.0%	10 – 50	29.5
MCG	Hard	100,000	50.0%	50.0%	20 – 80	48.2
LL	Synthetic	100,000	50.0%	50.0%	4 – 50	24.9
LL	Augmented	100,000	50.0%	50.0%	10 – 50	28.1
LL	Hard	100,000	50.0%	50.0%	20 – 80	45.7
KT	Synthetic	100,000	50.0%	50.0%	4 – 50	26.2
KT	Augmented	100,000	50.0%	50.0%	10 – 50	28.8
KT	Hard	100,000	50.0%	50.0%	20 – 80	47.3
SL2Z	Synthetic	100,000	50.0%	50.0%	4 – 50	26.5
SL2Z	Augmented	100,000	50.0%	50.0%	10 – 50	29.0
SL2Z	Hard	100,000	50.0%	50.0%	20 – 80	47.9

11.2 Grand Totals

Group	Full Name	Total Examples	Files
SC	Small Cancellation	300,000	3
TF	Thompson's F	300,000	3
H3	Heisenberg	300,000	3
BS12	Baumslag-Solitar	300,000	3
OR	One-Relator	300,000	3
GG	Grigorchuk	300,000	3
MCG	Mapping Class	300,000	3
LL	Lamplighter	300,000	3
KT	Knot (Trefoil)	300,000	3
SL2Z	Modular	300,000	3
TOTAL		**3,000,000**	**30**

12 Group Classification

12.1 By Algorithmic Properties

Property	Groups
$O(n)$ word problem	SC, F2
Polynomial word problem	TF, H3, OR, GG, MCG, LL, KT, SL2Z
Automatic	SC, TF, H3, OR, GG, MCG, LL, KT, SL2Z
Non-automatic	**BS12**

12.2 By Algebraic Structure

Structure	Groups
Free-like	SC (hyperbolic)
Nilpotent	H3 (2-step)
Solvable	LL (metabelian)
HNN extension	BS12
Self-similar	GG
Surface group related	MCG, TF
Knot group	KT
Arithmetic	SL2Z

12.3 By Special Features

Feature	Groups	Notes
Self-inverse generators	GG, LL	$a^2 = 1$ for some generators
Central element	H3	c commutes with all
Braid relations	MCG, KT	$\tau_i\tau_j\tau_i = \tau_j\tau_i\tau_j$
Torsion elements	SL2Z, GG	Finite order elements
Intermediate growth	GG	Between polynomial and exponential

12.4 Alphabet Comparison

Group	Alphabet Size	Token Format	Generators	Inverses
SC	4	chars	a, b	A, B
TF	4	space-sep	x0, x1	X0, X1
H3	6	space-sep	a, b, c	A, B, C
BS12	4	space-sep	a, t	A, T
OR	4	chars	a, b	A, B
GG	4	space-sep	a, b, c, d	(self-inverse)
MCG	10	space-sep	t1–t5	T1–T5
LL	3	space-sep	a, t	(a self-inv), T
KT	4	chars	a, b	A, B
SL2Z	4	space-sep	S, T	s, t

13 Critical Edge Cases for Neural Networks

13.1 Common Pitfalls by Group

Group	Trap Example	Equals	Why Tricky
H3	`a b A B`	c (not 1!)	Commutator relation
BS12	`t a T`	undefined	Would need $a^{1/2}$
GG	`b c`	d	Self-inverse with $bcd = 1$
LL	`t a T a`	$\neq 1$	Lamps at different positions
SL2Z	`S S`	$\neq 1$	S has order 4, not 2

13.2 Identity Patterns Summary

Group	Key Identity Pattern	Explanation
SC	Relator insertion	$ababab ABABAB = 1$
TF	Relation: $X_1 x_0 x_1 = x_0 x_0$	Tree pair normalization
H3	`a b A B C`	Commutator times C
BS12	`T a t A A`	$t^{-1}at = a^2$ applied
OR	`a b a b B A B A`	Single relator
GG	`b c d` or `a a`	Self-inverse + $bcd = 1$
MCG	`t1 t2 t1 T2 T1 T2`	Braid relation
LL	`t a t T a T`	Same lamp toggled twice
KT	`a b a B A B`	$aba = bab$ applied
SL2Z	`S T S T S T s s`	$(ST)^3 = S^2$